



HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering
Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT323
Midterm Examination
Date: 6/13/2026

Total Time: 1:30 Hours

Course Title: Numerical Analysis
Semester: Spring 2026
Total Marks: 30

$3 \times 10 = 30$

Answer any three (03) out of the following questions:

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No	Questions	CLOs	Marks												
Q1.	(a) Derive the Newton Raphson method.	CLO2	[5]												
	(b) Using the Newton Raphson method, find the root of the equation $x^3 - 2x - 5 = 0$, correct to four decimal places.	CLO3	[5]												
Q2.	(a) Using the bisection method, find the root of equation $x^3 - 4x + 1 = 0$ correct to four decimal places.	CLO3	[5]												
	(b) Derive the trapezoidal method.	CLO2	[5]												
Q3.	Using the trapezoidal and Simpson's $\frac{1}{3}$ rule, approximate the following definite integral $\int_1^2 x e^x dx$ using the same number of subintervals for both methods. Also compute the exact value of the integral and determine the absolute error for each method.	CLO3	[10]												
Q4.	(a) Using Newton's forward interpolation method, evaluate $f(x)$ at $x = 0.05$ using the following data. <table><tr><td>x</td><td>0.00</td><td>0.10</td><td>0.20</td><td>0.30</td><td>0.40</td></tr><tr><td>$y = f(x)$</td><td>1.000</td><td>1.2214</td><td>1.4918</td><td>1.8221</td><td>2.255</td></tr></table>	x	0.00	0.10	0.20	0.30	0.40	$y = f(x)$	1.000	1.2214	1.4918	1.8221	2.255	CLO3	[5]
x	0.00	0.10	0.20	0.30	0.40										
$y = f(x)$	1.000	1.2214	1.4918	1.8221	2.255										
	(b) The population (in thousand) of a town at different years is given in the following table. Using the Lagrange interpolation formula, evaluate the population at 2012. <table><tr><td>Year</td><td>2000</td><td>2005</td><td>2010</td><td>2015</td><td>2020</td></tr><tr><td>Population</td><td>50</td><td>65</td><td>80</td><td>92</td><td>105</td></tr></table>	Year	2000	2005	2010	2015	2020	Population	50	65	80	92	105	CLO3	[5]
Year	2000	2005	2010	2015	2020										
Population	50	65	80	92	105										
Q5.	(a) With the help of Newton's backward interpolation method approximate $f(-1/3)$ using the following information. <table><tr><td>x</td><td>-0.75</td><td>-0.50</td><td>-0.25</td><td>0</td></tr><tr><td>$y = f(x)$</td><td>-0.0718125</td><td>-0.02475</td><td>0.3349375</td><td>1.10100</td></tr></table>	x	-0.75	-0.50	-0.25	0	$y = f(x)$	-0.0718125	-0.02475	0.3349375	1.10100	CLO3	[6]		
x	-0.75	-0.50	-0.25	0											
$y = f(x)$	-0.0718125	-0.02475	0.3349375	1.10100											
	(b) Using the three-point forward difference formula and three-point central difference formula, approximate $f'(2)$ for $f(x) = e^x + 1$ with $h = 0.1$. Also compute the exact derivative at $x = 2$ and compute the error for each method.	CLO3	[4]												

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 321: Operating System

Batch: 25th

Date: 20.06.26

Midterm Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions.

$5 \times 6 = 30$

1. (a) What is the convey effect? CLO4 2
 (b) Write down the CPU scheduling criteria and describe them. CLO1 4

2. (a) Define context switch. CLO4 2
 (b) What is the problem with priority scheduling? What method can we apply to solve this problem? CLO4 4

3. (a)

Process	Burst Time
P1	5
P2	7
P3	9
P4	1
P5	5
P6	3

CLO4 3

Apply SJF, Calculate CT, TAT, WT, RT

- (b) Write down the advantages and disadvantages of the SJF algorithm. CLO4 3

4. (a)

Process	Arrival Time	Burst Time	Priority
P1	3	4	5
P2	3	10	1
P3	3	2	4
P4	3	11	6
P5	3	7	2
P6	3	3	3

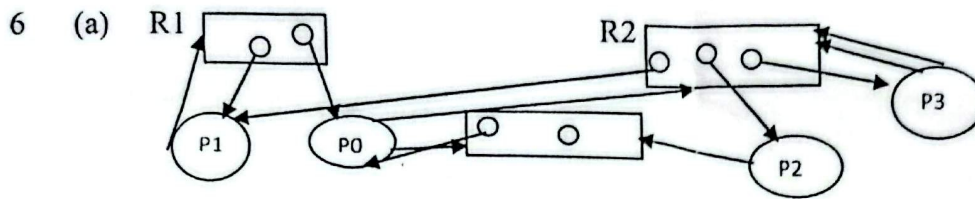
CLO4 3

Apply the priority scheduling algorithm, calculate CT, TAT, WT, RT.
 Assume minimum priority = P2

- (b) Write down Banker's algorithms. CLO3 3

5. (a) Describe Multi-level queue scheduling.
 (b) Describe deadlock. Write down the characteristics of deadlock.

CLO3 2
 CLO3 4



CLO3 3

R3

Apply the Banker's algorithm and check whether all the processes are in a safe state or not.

- (b) Write down the approaches to deadlock prevention.

CLO3 3

7. (a)

Process	Need [R0 R1 R2 R3 R4 R5]	Allocation [R0 R1 R2 R3 R4 R5]	Avilable
P0	[1 0 0 0 0 0]	[0 0 0 0 1 0]	[0 0 1 2 0 1]
P1	[0 0 1 0 1 0]	[0 1 0 0 0 0]	
P2	[0 0 0 0 0 1]	[0 0 1 0 1 0]	
P3	[0 0 0 0 0 0]	[1 1 0 1 0 1]	

CLO3 3

Apply Banker's algorithm and show the safe states sequence.

- (b) Apply the Round Robin(TQ=3) on the following table and calculate CT, TAT, WT, RT.

CLO4 3

Process	Arrival Time	Burst Time
P1	4	8
P2	10	2
P3	3	6
P4	5	1
P5	7	11
P6	3	5

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE325: Computer Peripherals and Interfacing

Batch: 25th

Date: 15/06/2026

Midterm Examination

Semester: Spring 2026

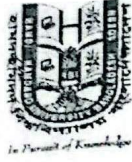
Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions.

5 × 6 = 30

- | | | |
|---|------|---|
| 1. (a) Define computer peripherals. Explain the term with proper context. | CLO1 | 2 |
| (b) Classify different types of peripherals and describe each type with suitable examples. | CLO1 | 4 |
| 2. (a) Describe the importance of peripherals in a computer system. | CLO1 | 3 |
| (b) Classify the keyboard based on the types of switches used in its keys. Explain each category with appropriate examples. | CLO1 | 3 |
| 3. (a) List the different groups of keys on a standard keyboard and briefly describe each group. | CLO1 | 2 |
| (b) Describe the working principle of the CRT monitor that uses an electron gun. | CLO1 | 4 |
| 4. (a) Mention and describe the common pins of memory devices. | CLO5 | 3 |
| (b) Draw and explain the timing diagram of the Intel-2716 EPROM. | CLO5 | 3 |
| 5. (a) Draw the pin diagram of an EPROM and describe all the pins. | CLO4 | 3 |
| (b) Mention and describe the operating modes of the Intel-2716 EPROM. | CLO4 | 3 |
| 6. (a) Describe the operation of a NAND-gate decoder with a suitable diagram. Determine the address range based on your diagram. | CLO5 | 3 |
| (b) Explain how a decoder can be used for address decoding when interfacing multiple ROM chips. Provide an appropriate diagram and specify the address ranges for the ROMs. | CLO5 | 3 |
| 7. Design a NAND gate decoder for Intel-2716 where the starting address for the microprocessor is (CF800) ₁₆ | CLO5 | 6 |



HAMBARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering & Technology
Department of Computer Science and Engineering
Program BSc

MGT 329: Industrial and Operational Management
Mid-Term Examination
Spring 2026

6th Semester

25th Batch
17. 06. 2026

Time: 1 Hour 30 Minutes

Total Marks: 30

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answer to questions should be neat and clean.

No.	Question/ Statement	CLOs	Marks																				
1.	Define Operations Management. Discuss the differences between goods and services using suitable examples. OR, Analyze how operations strategies contribute to business competitiveness.	1	6																				
2.	Explain the concept of productivity and discuss the factors affecting productivity. OR, A company produced 2,500 units using 500 labor hours. Calculate the labor productivity. If output increases to 3,000 units with the same labor hours, determine the new productivity and compare the results.	1	5																				
3.	Explain the steps in the forecasting process. OR, Explain different process types with suitable examples.	2,3	5																				
4.	Given the following weekly demand data, determine the exponential smoothing forecast for Week 10 using $\alpha = 0.10$. Assume that $F_1 = D_1$. <table border="1"><tr><th>Week</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th><th>9</th></tr><tr><td>Demand</td><td>820</td><td>775</td><td>680</td><td>655</td><td>750</td><td>802</td><td>798</td><td>689</td><td>775</td></tr></table> OR, Compute the five-year moving average from the following data.	Week	1	2	3	4	5	6	7	8	9	Demand	820	775	680	655	750	802	798	689	775	2	6
Week	1	2	3	4	5	6	7	8	9														
Demand	820	775	680	655	750	802	798	689	775														

Year	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991
Values	1114	1117	1119	1126	1137	1140	1142	1145	1150	1155

5

The following table shows the actual weekly sales for 10 weeks:

Week (t)	1	2	3	4	5	6	7	8	9	10
Actual Sale (y)	700	724	720	728	740	742	758	750	770	775

Using linear trend equation, Forecast the sales for week 11 and week 12.

OR,

Use the information in the table below to balance the assembly line for Deli Pizza Company. The operations run continuously for 8 hours per day, and 200 pizzas must be made to meet demand.

Table: Task Information for Assembly Line

Activity	Task Description	Predecessors	Time (Minutes)
A	Shape the dough to form the crust	None	2.00
B	Add pizza sauce	A	1.00
C	Add cheese	B	2.00
D	Add sausage	C	0.75
E	Add pepperoni	C	1.00
F	Package the pizza	D, E	1.50
G	Label the package	F	0.50
Total task time (Σt) = 8.75 minutes			

Required Tasks:

1. Draw the precedence diagram of the production process.
2. Balance the assembly line using the Longest Task Time Rule.

Calculate the efficiency of the balanced line.

2, 3

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Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE323: System Analysis & Design

Batch: 25th
Date: 10-06-2026

Mid-Term Examination
Semester: Spring 2026

Time: 1 Hour 30 Minutes
Total Marks: 30

Answer any **five** questions from the followings.

5 × 6 = 30

		CLOs	Marks																					
1.	(a) Who are the typical stakeholders of an Information System? Describe them with example.	CLO1	3																					
	(b) Summarize different types of users of a system.	CLO1	3																					
2.	(a) Why is it important for a systems analyst to interact with users and management from the beginning of a project?	CLO2	3																					
	(b) How can feedback from employees help in identifying organizational problems?	CLO2	3																					
	(c) Why is external feedback important for identifying organizational problems?	CLO2																						
3.	(a) Define the term Organizational Boundary .	CLO1	2																					
	(b) What is the purpose of defining problems and objectives in a system? How does it guide the systems analyst?	CLO1	4																					
4.	(a) Define an entity and a relationship in the context of an entity-relationship (E-R) model. Provide an example.	CLO1	3																					
	(b) Explain different types of feasibility analysis needed before the initiation of a system.	CLO1	3																					
5.	(a) What is the purpose of using analogies when estimating project time?	CLO1	2																					
	(b) Using the data provided in the tables for costs and benefits over six years, Given a discount rate of 11%, calculate the present value of the Year-3 cost	CLO3	4																					
<table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Costs</td><td>35000</td><td>40000</td><td>45000</td><td>48000</td><td>52000</td><td>55000</td></tr><tr><td>Benefits</td><td>20000</td><td>28000</td><td>50000</td><td>60000</td><td>75000</td><td>84000</td></tr></table>				Year	1	2	3	4	5	6	Costs	35000	40000	45000	48000	52000	55000	Benefits	20000	28000	50000	60000	75000	84000
Year	1	2	3	4	5	6																		
Costs	35000	40000	45000	48000	52000	55000																		
Benefits	20000	28000	50000	60000	75000	84000																		
6.	(a) If an organization does not have an updated computer hardware inventory, what essential information should a systems analyst collect while setting up a new hardware inventory system?	CLO1	3																					
	(b) When evaluating vendor support for computer hardware, what major service categories should an organization consider, and what specific support offerings typically fall under each category? Explain briefly.	CLO1	3																					

7. (a) What are the main benefits of BYOD? CLO1 2
 (b) A frozen-food company wants to develop an information system for tracking shipments to warehouses. The information of the development process is given below: CLO4 4

Task	Must Follow	Time (Weeks)
A	None	5
B	None	3
C	B	2
D	A	3
E	C, D	4
F	E	4
G	E	5
H	F	2
I	G	3
J	I	6

Draw PERT diagram for the above system and identify the critical path.